Counting Polygon and Polygon Free Mosaics

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Abstract

Imagine you are tiling a rectangular floor with seven distinct types of square tiles of unit length. Six of the tile types have a line segment connecting a pair of sides at their midpoints. The seventh tile is the "blank" tile with no line segment. Some floors covered in these tiles, referred to as mosaics in the literature, can contain polygons formed by the line segments. Hong and Oh gave bounds on the number of mosaics in which every tile is either part of a polygon or the blank tile, and we instead give an exact count of these so called polygon mosaics. We also introduce and solve the more difficult problem of counting polygon free mosaics, which are mosaics that contain no polygons.

1 Mosaics

Imagine you are tiling a rectangular floor that is m units by n units, where m and n are positive integers. At your disposal is an unending supply of seven distinct types of square tiles of unit length, which can include an additional dotted line segment, as in Figure 1. The first tile type is the "blank tile" with no additional line segment, and the other six tile types each has a line segment that connects, in all possible ways, the midpoints of two sides.

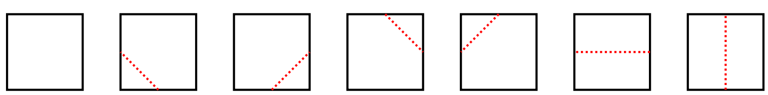
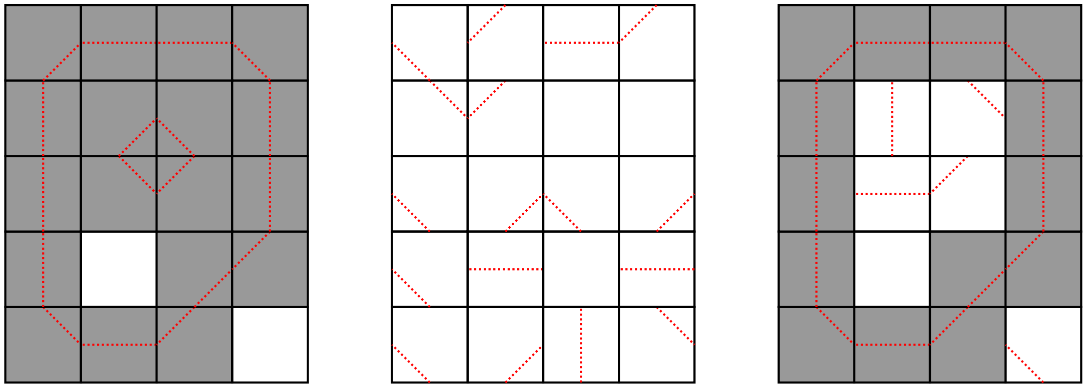


Figure 1: The tile set

The task is complete once you cover the floor, which requires mn tiles. An (m, n) *mosaic* is a fully tiled floor with m rows and n columns. Figure 2 shows various $(5, 4)$ mosaics, with some added shading. Notice that the segments from some tiles form polygons and others do not; in Figure 2 we shade the tiles that are part of any polygon. Our goal in this paper is, for any given m and n, to count the number of polygon mosaics and the number of polygon free mosaics.

Definition 1.1. An (m, n) mosaic is a *polygon mosaic* if all line segments are part of polygons, and it is a *polygon free mosaic* if the line segments form no polygons.



(a) A polygon mosaic

(b) A polygon free mosaic

(c) A mosaic that is neither a polygon mosaic nor a polygon free mosaic

Figure 2: Examples of $(5, 4)$ mosaics, with tiles forming polygons shaded

Figure 2a depicts a polygon mosaic, Figure 2b depicts a polygon free mosaic, and Figure 2c depicts a mosaic that is neither a polygon free mosaics nor a polygon mosaic. Notice that all tiles are blank in any polygon mosaic that is polygon free.

The types of polygons that can be formed from the tiles are more commonly known as "self-avoiding polygons" in the literature [2, 3] to highlight their connection with self-avoiding walks. As self-avoiding polygons are formed by walks on a square lattice, some authors [4] have chosen to represent the tiles connecting adjacent sides with two line segments that meet at the center of the square at a right angle, as in Figure 3. The results of this paper are true for either style of tile.

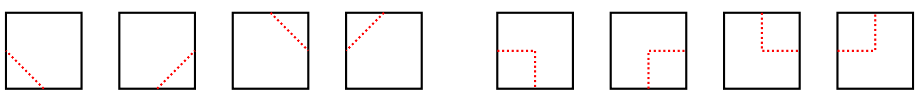


Figure 3: Our tile style (left) and another common style [4] (right)

To begin, we introduce a bijection between polygon mosaics and closely related objects we call proper lattices.

2 Mapping Polygon Mosaics to Proper Lattices

For positive integers m and n, an (m, n) *lattice* is a rectangular grid that has $m+1$ rows and $n+1$ columns of integers, which we call *vertices*. All our lattices will be *binary*, meaning that each vertex is 0 or 1. A *cell* is a $(1, 1)$ binary lattice, so that an (m, n) binary lattice is made up of mn cells. The sixteen possible cells are

$$\begin{smallmatrix}0&0\\0&0\end{smallmatrix},\ \begin{smallmatrix}0&0\\0&1\end{smallmatrix},\ \begin{smallmatrix}0&0\\1&0\end{smallmatrix},\ \begin{smallmatrix}0&0\\1&1\end{smallmatrix},\ \begin{smallmatrix}0&1\\0&0\end{smallmatrix},\ \begin{smallmatrix}0&1\\0&1\end{smallmatrix},\ \begin{smallmatrix}0&1\\1&0\end{smallmatrix},\ \begin{smallmatrix}0&1\\1&1\end{smallmatrix},\ \begin{smallmatrix}1&0\\0&0\end{smallmatrix},\ \begin{smallmatrix}1&0\\0&1\end{smallmatrix},\ \begin{smallmatrix}1&0\\1&0\end{smallmatrix},\ \begin{smallmatrix}1&0\\1&1\end{smallmatrix},\ \begin{smallmatrix}1&1\\0&0\end{smallmatrix},\ \begin{smallmatrix}1&1\\0&1\end{smallmatrix},\ \begin{smallmatrix}1&1\\1&0\end{smallmatrix},\ \text{and}\ \begin{smallmatrix}1&1\\1&1\end{smallmatrix}.$$

We further define that a lattice is *framed* if it is binary and every vertex on the boundary of the lattice is 0, and a lattice is *proper* if it is framed and does not contain any $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$ or $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$ cells. For example, the right side of Figure 4 is a $(5,4)$ proper lattice.

Let P be a polygon mosaic. We will create a binary lattice from P by replacing every point that is at a corner of a tile of P by either 0 or 1. For each such point, we count the number of polygons of P that surround it, i.e., polygons for which the point is in the polygon's interior. If this number is even, label the vertex 0, and if it is odd, label the vertex 1. We then delete all tile edges and dotted line segments, and define $\mathscr{L}(P)$ to be the resulting binary lattice. See Figure 4 for an example. If P is an (m,n) polygon mosaic, then $\mathscr{L}(P)$ is an (m,n) binary lattice. In fact, $\mathscr{L}(P)$ is an (m,n) framed lattice because every point on the edge of P is surrounded by no polygons.

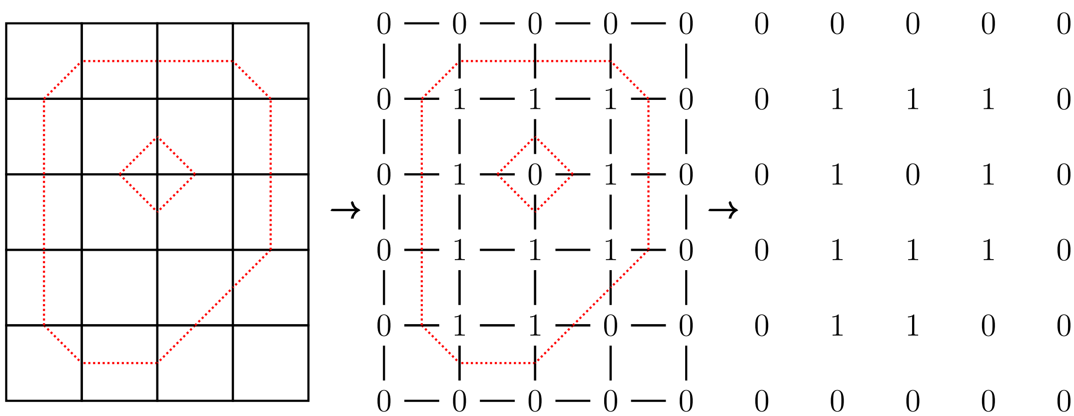


Figure 4: A polygon mosaic P (left), its image $\mathscr{L}(P)$ (right), and the two superimposed (middle).

In our definition of $\mathscr{L}$, a tile of a polygon mosaic P corresponds to a cell of $\mathscr{L}(P)$. It can be helpful to overlay P and $\mathscr{L}(P)$, such as in the middle of Figure 4. Whenever we do this, in any tile that is not blank, the dotted line segment separates vertices labeled 0 on one side and labeled 1 on the other. Any given tile can map to one of two cells. For example, the tile second from left in Figure 1 must map to either the cell $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$ or the cell $\begin{smallmatrix}1&1\\0&1\end{smallmatrix}$. Which one? Look first where the interior of this polygon is relative to this dotted line segment, and second, the number of other polygons that contain this polygon. For example, the tile maps to $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$ if the interior is below the dotted line segment and an even number of polygons contain this polygon. A blank tile of P, on the other hand, corresponds to a cell that is either $\begin{smallmatrix}0&0\\0&0\end{smallmatrix}$ or $\begin{smallmatrix}1&1\\1&1\end{smallmatrix}$. With seven tiles, and no cell able to be the image of distinct tiles, $\mathscr{L}(P)$ contains up to fourteen types of cells. The two types that never appear in $\mathscr{L}(P)$ are $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$ and $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$. Because $\mathscr{L}(P)$ is also framed, $\mathscr{L}(P)$ is thus proper.

Given any proper lattice ℓ, we can reverse this process, creating a mosaic by replacing each cell by the unique tile that can map to it. Define $\mathscr{L}^{-1}$ by restricting this map to proper lattices, and let us show that $\mathscr{L}^{-1}$ is indeed an inverse to $\mathscr{L}$. This will prove that $\mathscr{L}$ gives a bijection from (m,n) polygon mosaics to (m,n) proper lattices. Our discussion above shows that $\mathscr{L}^{-1}\circ\mathscr{L}(P)=P$ for any polygon mosaic P, so fixing a proper lattice ℓ, we must show that $\mathscr{L}^{-1}(\ell)$ is a polygon mosaic and that $\mathscr{L}\circ\mathscr{L}^{-1}(\ell)=\ell$.

First observe that polygon mosaics can be characterized as those mosaics in which every end of a dotted line segment on a tile meets, on an adjacent tile, another end of a dotted line segment. In the mosaic $\mathscr{L}^{-1}(\ell)$, these ends are exactly the points, in the corresponding cell of ℓ, which lie halfway between adjacent vertices of ℓ, one labeled 0 and the other labeled 1. Because ℓ is a framed lattice, the vertex labeled 1 is an internal vertex. Thus, another cell of ℓ shares these two vertices, and the dotted lines of $\mathscr{L}^{-1}(\ell)$ from the two corresponding tiles meet. Therefore $\mathscr{L}^{-1}(\ell)$ is a polygon mosaic.

By our analysis of $\mathscr{L}$ and our definition of $\mathscr{L}^{-1}$, each cell of $\mathscr{L} \circ \mathscr{L}^{-1}(\ell)$ must either be equal to the corresponding cell of ℓ or is formed from the corresponding cell by relabeling all vertices 0 as 1 and 1 as 0. Furthermore, since the cells are connected, i.e., none is isolated, we must then have that $\mathscr{L} \circ \mathscr{L}^{-1}(\ell)$ itself is either equal to ℓ or formed by relabeling all vertices 0 as 1 and 1 as 0. By hypothesis, ℓ is a proper lattice, and thus a framed lattice, and by construction, every binary lattice in the range of $\mathscr{L}$ is a framed lattice. Therefore $\mathscr{L} \circ \mathscr{L}^{-1}(\ell) = \ell$. This concludes our proof that $\mathscr{L}^{-1}$ is the inverse of $\mathscr{L}$ and thus that $\mathscr{L}$ is bijective.

3 Counting Polygon Mosaics

Let us now introduce certain functions that allow us to count both polygon mosaics (in this section) and polygon free mosaics (Section 4), as well as certain variations we will create by varying the tile set (Section 6).

Definition. A *cell product function* is a real valued function whose domain is the set of all binary lattices and whose value on any lattice is the product of its values on every cell in the lattice.

Theorem 3.1. *Let v be the cell product function that maps the cells $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$ and $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$ to 0, and maps the fourteen other cells cells to 1. Then for all positive integers m and n, the number of (m,n) polygon mosaics equals $\sum_\ell v(\ell)$, where the sum is over all (m,n) framed lattices ℓ.*

Proof. For any framed lattice ℓ, $v(\ell)$ is 1 if ℓ is proper, and otherwise is 0, so that $\sum_\ell v(\ell)$ counts the number of (m,n) proper lattices. Our conclusion now follows from the fact that $\mathscr{L}$ gives a bijection from the set of (m,n) polygon mosaics to the set of (m,n) proper lattices. $\square$

For any cell product function v, we can use matrices to efficiently compute the sum $\sum_\ell v(\ell)$, where the sum is over all (m,n) framed lattices ℓ.

Theorem 3.2. *Let v be any cell product function. Define the 1×1 matrices $A_1 = \left(v\left(\begin{smallmatrix}0&0\\0&0\end{smallmatrix}\right)\right)$, $B_1 = \left(v\left(\begin{smallmatrix}0&0\\1&0\end{smallmatrix}\right)\right)$, $C_1 = \left(v\left(\begin{smallmatrix}1&0\\0&0\end{smallmatrix}\right)\right)$, and $D_1 = \left(v\left(\begin{smallmatrix}1&0\\1&0\end{smallmatrix}\right)\right)$, and for positive integers n, define the $2^n \times 2^n$ matrices*

$$A_{n+1} = \begin{pmatrix} v(\begin{smallmatrix}0&0\\0&0\end{smallmatrix})A_n & v(\begin{smallmatrix}0&0\\0&1\end{smallmatrix})B_n \\ v(\begin{smallmatrix}0&1\\0&0\end{smallmatrix})C_n & v(\begin{smallmatrix}0&1\\0&1\end{smallmatrix})D_n \end{pmatrix} \quad B_{n+1} = \begin{pmatrix} v(\begin{smallmatrix}0&0\\1&0\end{smallmatrix})A_n & v(\begin{smallmatrix}0&0\\1&1\end{smallmatrix})B_n \\ v(\begin{smallmatrix}0&1\\1&0\end{smallmatrix})C_n & v(\begin{smallmatrix}0&1\\1&1\end{smallmatrix})D_n \end{pmatrix}$$
$$C_{n+1} = \begin{pmatrix} v(\begin{smallmatrix}1&0\\0&0\end{smallmatrix})A_n & v(\begin{smallmatrix}1&0\\0&1\end{smallmatrix})B_n \\ v(\begin{smallmatrix}1&1\\0&0\end{smallmatrix})C_n & v(\begin{smallmatrix}1&1\\0&1\end{smallmatrix})D_n \end{pmatrix} \quad D_{n+1} = \begin{pmatrix} v(\begin{smallmatrix}1&0\\1&0\end{smallmatrix})A_n & v(\begin{smallmatrix}1&0\\1&1\end{smallmatrix})B_n \\ v(\begin{smallmatrix}1&1\\1&0\end{smallmatrix})C_n & v(\begin{smallmatrix}1&1\\1&1\end{smallmatrix})D_n \end{pmatrix}.$$

Then for all positive integers m and n, the upper left entry of A_n^m equals $\sum_\ell v(\ell)$, where the sum is over all (m,n) framed lattices ℓ.

In the statement of Theorem 3.2, we evaluate v at each of the sixteen possible cells at least once. The order in which these values appear may look haphazard but is exactly what we need to establish the first part of Proposition 3.3, which we will use to prove the proposition's second part, which in turn is a key part of the proof of Theorem 3.2.

Until we complete the proof of Theorem 3.2, we use, as found there, v and the matrices A_n, B_n, C_n, and D_n, for $n \geq 1$. Also, let $\beta_n(k)$ be the n digit binary representation of the nonnegative integer $k < 2^n$, separated out into individual digits (e.g., $\beta_3(5)$ is 1 0 1 rather than 101), with $\beta_0(0)$ being the empty string. This allows us, for example, to write the binary lattice $\begin{smallmatrix} 0&1&1&0 \\ 0&0&1&0 \end{smallmatrix}$ as $\begin{smallmatrix} \beta_4(6) \\ \beta_4(2) \end{smallmatrix}$ or as $\begin{smallmatrix} 0 & \beta_2(3) & 0 \\ 0 & \beta_2(1) & 0 \end{smallmatrix}$. Additionally, we number rows and columns of all matrices beginning at zero.

Proposition 3.3. *Let n be a positive integer. Numbering rows and columns beginning at 0, we have the following.*

1. $\begin{pmatrix} A_n & B_n \\ C_n & D_n \end{pmatrix}$ *equals the $2^n \times 2^n$ matrix whose (i,j) entry is* $v\left(\begin{smallmatrix} \beta_{n+1}(2i) \\ \beta_{n+1}(2j) \end{smallmatrix}\right)$.
2. A_n *equals the $2^{n-1} \times 2^{n-1}$ matrix whose (i,j) entry is* $v\left(\begin{smallmatrix} 0 & \beta_{n-1}(i) & 0 \\ 0 & \beta_{n-1}(j) & 0 \end{smallmatrix}\right)$.

Proof. Notice that for all nonnegative integers $k < 2^n$, $\beta_{n+1}(2k)$ is the string $\beta_n(k)\ 0$ and that if $k < 2^{n-1}$ then $\beta_n(k)$ is the string $0\ \beta_{n-1}(k)$. Putting these together, the first part of the proposition implies the second.

Define X_n to be the $2^n \times 2^n$ matrix $\begin{pmatrix} A_n & B_n \\ C_n & D_n \end{pmatrix}$ and Y_n to be the $2^n \times 2^n$ matrix whose (i,j) entry is $v\left(\begin{smallmatrix} \beta_{n+1}(2i) \\ \beta_{n+1}(2j) \end{smallmatrix}\right)$. To complete the proposition, we must prove $X_n = Y_n$ for all positive integers n. Since $\beta_2(0)$ is 0 0 and $\beta_2(2)$ is 1 0, the $n = 1$ case follows from the definitions of A_1, B_1, C_1, and D_1 in Theorem 3.2. Continuing by induction, we assume $X_n = Y_n$ and prove $X_{n+1} = Y_{n+1}$.

The statement of Theorem 3.2 tells us that

$$X_{n+1} = \begin{pmatrix} v\left(\begin{smallmatrix} 0&0 \\ 0&0 \end{smallmatrix}\right) A_n & v\left(\begin{smallmatrix} 0&0 \\ 0&1 \end{smallmatrix}\right) B_n & v\left(\begin{smallmatrix} 0&0 \\ 1&0 \end{smallmatrix}\right) A_n & v\left(\begin{smallmatrix} 0&0 \\ 1&1 \end{smallmatrix}\right) B_n \\ v\left(\begin{smallmatrix} 0&1 \\ 0&0 \end{smallmatrix}\right) C_n & v\left(\begin{smallmatrix} 0&1 \\ 0&1 \end{smallmatrix}\right) D_n & v\left(\begin{smallmatrix} 0&1 \\ 1&0 \end{smallmatrix}\right) C_n & v\left(\begin{smallmatrix} 0&1 \\ 1&1 \end{smallmatrix}\right) D_n \\ v\left(\begin{smallmatrix} 1&0 \\ 0&0 \end{smallmatrix}\right) A_n & v\left(\begin{smallmatrix} 1&0 \\ 0&1 \end{smallmatrix}\right) B_n & v\left(\begin{smallmatrix} 1&0 \\ 1&0 \end{smallmatrix}\right) A_n & v\left(\begin{smallmatrix} 1&0 \\ 1&1 \end{smallmatrix}\right) B_n \\ v\left(\begin{smallmatrix} 1&1 \\ 0&0 \end{smallmatrix}\right) C_n & v\left(\begin{smallmatrix} 1&1 \\ 0&1 \end{smallmatrix}\right) D_n & v\left(\begin{smallmatrix} 1&1 \\ 1&0 \end{smallmatrix}\right) C_n & v\left(\begin{smallmatrix} 1&1 \\ 1&1 \end{smallmatrix}\right) D_n \end{pmatrix},$$

so that each entry of X_{n+1} is v evaluated at a cell times an entry of A_n, B_n, C_n, or D_n. After we also express every entry of Y_{n+1} as v evaluated at a cell times another quantity, we will show that the cells evaluated at v are the same, as are the other quantities, and thus $X_{n+1} = Y_{n+1}$. To express an entry of Y_{n+1} in our desired way, we use that v is a cell product function to write $v\left(\begin{smallmatrix} \beta_{n+2}(2i) \\ \beta_{n+2}(2j) \end{smallmatrix}\right)$ as the product of v applied to the leftmost cell of $\begin{smallmatrix} \beta_{n+2}(2i) \\ \beta_{n+2}(2j) \end{smallmatrix}$ and $v(\ell_{i,j})$, where $\ell_{i,j}$ is the binary lattice formed by deleting the leftmost column of $\begin{smallmatrix} \beta_{n+2}(2i) \\ \beta_{n+2}(2j) \end{smallmatrix}$.

Consider $\beta_{n+2}(2k)$ for $k = 0, 1, \ldots, 2^{n+1} - 1$. The first half of these 2^{n+1} strings begin with 0 and the second half begin with 1. Splitting each half in half, the strings in each of these four parts begin with 0 0, 0 1, 1 0, and 1 1, in that order. Therefore, as we go down any column of Y_{n+1} from the top row to the bottom row in four equal parts, the leftmost cell of $\substack{\beta_{n+2}(2i)\\ \beta_{n+2}(2j)}$ has upper row 0 0, 0 1, 1 0, then 1 1. Similarly, going to the right across any row of Y_{n+1}, the leftmost cell has lower row 0 0, 0 1, 1 0, then 1 1. Comparing this to X_{n+1} as above, the cells evaluated by v are the same in both X_{n+1} and Y_{n+1}.

To complete our proposition, we must show for all i and j that $v(\ell_{i,j})$ is equal to the entry of A_n, B_n, C_n, or D_n that appears in the appropriate block of X_{n+1} as shown above. Returning to $\beta_{n+2}(2k)$ for $k = 0, 1, \ldots, 2^{n+1}-1$ and deleting the first digit of each string, this sequence becomes two copies of the sequence of strings given by $\beta_{n+1}(2k)$ for $k = 0, 1, \ldots, 2^n - 1$. Using this, consider the $2^{n+1} \times 2^{n+1}$ matrix whose (i,j) entry is $v(\ell_{i,j})$. As we go down any column, we get two copies of $\beta_{n+1}(2i)$ for $i = 0, 1, \ldots, 2^n - 1$ in the upper row of $\ell_{i,j}$, and as we move right across any row, we get two copies of $\beta_{n+1}(2j)$ for $j = 0, 1, \ldots, 2^n - 1$ in the lower row of $\ell_{i,j}$. Therefore, using the definition of Y_n, the matrix of $v(\ell_{i,j})$ is exactly $\begin{pmatrix} Y_n & Y_n \\ Y_n & Y_n \end{pmatrix}$. By our induction hypothesis, $Y_n = X_n = \begin{pmatrix} A_n & B_n \\ C_n & D_n \end{pmatrix}$, and thus in the (i,j) entry of X_{n+1} as given above, the entry of A_n, B_n, C_n, or D_n equals $v(\ell_{i,j})$, as needed. □

Proof of Theorem 3.2. Fix n. We first introduce some terminology. For all positive integers m, we say a *height m strip* is an (m,n) binary lattice in which all vertices in the farthest left and farthest right columns are labeled zero. Then for a height m strip ℓ and a nonnegative integer $i < 2^{n-1}$, we say a particular row of ℓ is *determined by i* if that row is equal to $0\ \beta_{n-1}(i)\ 0$. Note that an (m,n) framed lattice is exactly a height m strip in which both the top and bottom rows are determined by 0. Thus, to prove our theorem it suffices to show the following.

Claim. For all positive integers m and nonnegative integers $i, j < 2^{n-1}$, the (i,j) entry of A_n^m equals $\sum_\ell v(\ell)$, where the sum is over all height m strips ℓ whose first row is determined by i and whose last row is determined by j.

To prove the claim, first observe that for each i and j, there exists a unique height 1 strip whose first row is determined by i and whose last row is determined by j. By the second part of Proposition 3.3, v applied to this strip is equal to the (i,j) entry of A_n. This proves our claim when $m = 1$. We continue by induction, so for some $m \geq 2$, assume the claim for $m - 1$.

Fix nonnegative integers $i, j < 2^{n-1}$, and consider some nonnegative integer $k < 2^{n-1}$. As in the $m = 1$ case, the second part of Proposition 3.3 tells us that the (i,k) entry of A_n equals v applied to the unique height 1 strip with first and last rows determined respectively by i and by k. By our induction hypothesis, the (k,j) entry of A_n^{m-1} equals $\sum_\ell v(\ell)$, where the sum is over all height $m-1$ strips ℓ whose first and last rows are determined respectively by k and by j. Consider applying v both to the height 1 strip indicated above and to one of the height $m-1$ strips ℓ in this sum. Because v is a cell product function, the product of these two numbers is equal to v applied to the height m strip formed by overlapping the bottom row of the height 1 strip to the top row of ℓ, which we illustrate as

$$v\begin{pmatrix} 0 & \beta_{n-1}(i) & 0 \\ 0 & \beta_{n-1}(k) & 0 \end{pmatrix} v\begin{pmatrix} 0 & \beta_{n-1}(k) & 0 \\ \vdots & \vdots & \vdots \\ 0 & \beta_{n-1}(j) & 0 \end{pmatrix} = v\begin{pmatrix} 0 & \beta_{n-1}(i) & 0 \\ 0 & \beta_{n-1}(k) & 0 \\ \vdots & \vdots & \vdots \\ 0 & \beta_{n-1}(j) & 0 \end{pmatrix}.$$

From this we see that the product of the (i,k) entry of A_n and the (k,j) entry of A_n^{m-1} is equal to $\sum_\ell v(\ell)$, where the sum is now over all height m strips ℓ whose first, second, and last rows are determined respectively by i, by k, and by j. This tells us, in turn, that when we additionally sum over all nonnegative $k < 2^{n-1}$, we can conclude that the (i,j) entry of $A_n \cdot A_n^{m-1} = A_n^m$ is, as desired, the sum $\sum_\ell v(\ell)$ where the sum is over all height m strips ℓ whose first and last rows are determined respectively by i and by j. $\square$

Substituting the values for v from Theorem 3.1 into the matrices from Theorem 3.2 gives the following.

Theorem 3.4. *Define the* 1×1 *matrices* $A_1 = B_1 = C_1 = D_1 = \begin{pmatrix}1\end{pmatrix}$, *and for positive integers* n, *define the* $2^n \times 2^n$ *matrices*

$$A_{n+1} = \begin{pmatrix} A_n & B_n \\ C_n & D_n \end{pmatrix} \quad B_{n+1} = \begin{pmatrix} A_n & B_n \\ 0C_n & D_n \end{pmatrix}$$
$$C_{n+1} = \begin{pmatrix} A_n & 0B_n \\ C_n & D_n \end{pmatrix} \quad D_{n+1} = \begin{pmatrix} A_n & B_n \\ C_n & D_n \end{pmatrix}.$$

Then for all positive integers m *and* n, *the number of* (m,n) *polygon mosaics is the upper left entry of* A_n^m.

Now that we have counted polygon mosaics, we move to polygon free mosaics.

4 Counting Polygon Free Mosaics

The main goal of this section is to prove the following theorem.

Theorem 4.1. *Let* v *be the cell product function that maps the cells* $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$ *and* $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$ *to* 0, $\begin{smallmatrix}0&0\\0&0\end{smallmatrix}$ *and* $\begin{smallmatrix}1&1\\1&1\end{smallmatrix}$ *to* 7, $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$ *and* $\begin{smallmatrix}1&0\\1&1\end{smallmatrix}$ *to* -1, *and the ten other cells to* 1. *Then for all positive integers* m *and* n, *the number of* (m,n) *polygon free mosaics equals* $\sum_\ell v(\ell)$, *where the sum is over all* (m,n) *framed lattices* ℓ.

Throughout this section, let v be as in Theorem 4.1. For example, if ℓ is the proper lattice on the right of Figure 4, we have $v(\ell) = 49$, as there is one $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$ cell, one $\begin{smallmatrix}1&0\\1&1\end{smallmatrix}$ cell, one $\begin{smallmatrix}0&0\\0&0\end{smallmatrix}$ cell, and one $\begin{smallmatrix}1&1\\1&1\end{smallmatrix}$ cell. For the proof of Theorem 4.1, we need to following key result.

Proposition 4.2. *For a polygon mosaic* P, *let* $\ell = \mathscr{L}(P)$. *If* P *contains an even [resp. odd] number of polygons then* $v(\ell)$ *is positive [resp. negative].*

We might want to think of this result as relating a "global" characteristic and "local" characteristic, since the number of polygons in P is determined by looking at the entire mosaic, whereas $v(\ell)$ is calculated by looking at one cell at a time. The reader may wish to compare this to $\mathscr{L}$, which was defined "globally," versus $\mathscr{L}^{-1}$, which was defined "locally."

Proof of Proposition 4.2. Let N be the number of vertices of ℓ that are 1, and define ℓ_0 to be the lattice formed by changing these N vertices from 1 to 0. That is, ℓ_0 has the same dimensions as ℓ, but all its vertices are 0. We will change ℓ_0 back to ℓ by flipping these vertices back from 0 to 1, one vertex at a time. The order in which we flip vertices is as follows.

We flip vertices one row at a time, working from top to bottom. In each row we make two passes, each time moving from left to right. In both passes, the flipped vertex must be at the same place as a vertex of ℓ that is 1. In the first pass we additionally require that the vertex directly above is 1 (this vertex is 1 in both our current lattice and also in ℓ, since we are working the rows from top to bottom), whereas in the second pass the vertex directly above must be 0.

This procedure creates a sequence $\ell_0, \ell_1, \ldots, \ell_N$ of lattices with $\ell_N = \ell$. Because $\ell = \mathscr{L}(P)$, we know ℓ is a proper lattice. Since every ℓ_i has the same dimensions as ℓ and has vertices of 0 in at least the same positions as in ℓ, we see every ℓ_i is a framed lattice. We will prove that every ℓ_i is actually a proper lattice (which is the reason we make two passes through each row) and that by taking $P_i = \mathscr{L}^{-1}(\ell_i)$ for all i, the proposition holds for all P_i and ℓ_i. Since $\ell = \ell_N$, this will complete our proof.

In the transition from ℓ_{i-1} to ℓ_i, because every ℓ_i is framed, the single vertex that changes is adjacent to eight vertices, as in Figure 5, where the changing central vertex $\#$ is 0 in ℓ_{i-1} and 1 in ℓ_i, and the remaining vertices are labeled for their relative position to this central vertex, such as dl for down left and u for up.

$$\begin{array}{ccccc} ul & — & u & — & ur \\ | & & | & & | \\ l & — & \# & — & r \\ | & & | & & | \\ dl & — & d & — & dr \end{array}$$

Figure 5: Cases Diagram

Let us determine the adjacent vertices that can occur. Since our process goes from top to bottom one row at a time, we have $(dl, d, dr) = (0, 0, 0)$. Also, because of the way in which we make our two passes, if $u = 1$ then $r = 0$ and $(ul, l) \neq (0, 1)$, whereas if $u = 0$ then $(ur, r) \neq (0, 1)$. Additionally, in the case $u = 0$, neither (ul, l) nor (ur, r) equals $(1, 0)$. Indeed, if $(ul, l) = (1, 0)$ then we would have a cell $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$ in ℓ_i, and thus also in $\ell_N = \ell$, since we are making the second pass through the row. This would contradict that ℓ is a proper lattice. Similarly, $(ur, r) = (1, 0)$ implies that ℓ has a cell $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$ (since with $ur = 1$, r cannot change in the second pass), again contradicting that ℓ is a proper lattice. Putting all this together, on the first pass, i.e., when $u = 1$, the vertices can only be as in the six cases pictured in Figure 6. Similarly, on the second pass we have $u = 0$ and one of the six cases of Figure 7.

Since trivially ℓ_0 is a proper lattice, the twelve cases in Figures 6 and 7 show inductively that every ℓ_i is a proper lattice because we have already seen that every ℓ_i is framed, and none of the cases create a $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$ or $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$ cell. For every i, let $P_i = \mathscr{L}^{-1}(\ell_i)$. Figures 6 and 7 also show the portions of P_{i-1} and P_i that are in these four cells, where the dashed lines are

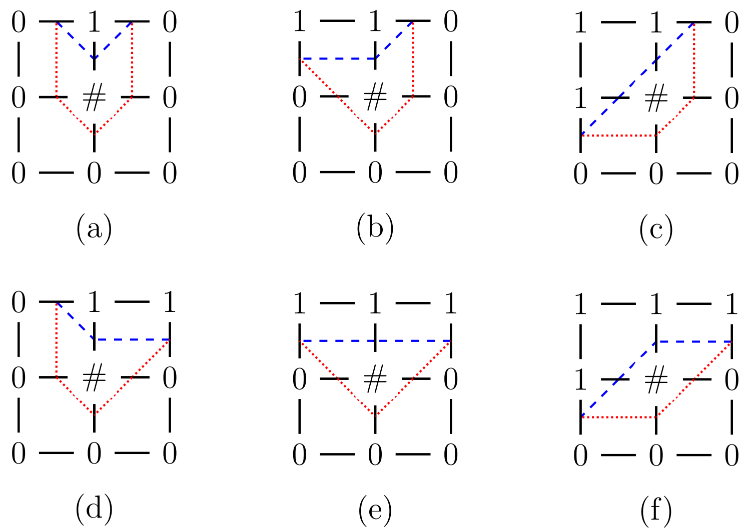


Figure 6: The cases when $u = 1$

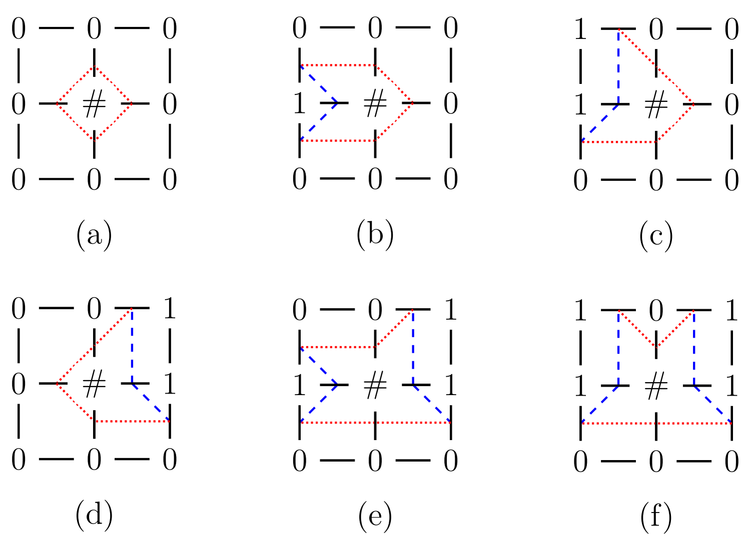


Figure 7: The cases when $u = 0$

from the polygons of P_{i-1} and the dotted lines from the polygons of P_i. (Recall that # is 0 in ℓ_{i-1} and 1 in ℓ_i.)

Let us say a cell is *negative* if it is $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$ or $\begin{smallmatrix}1&0\\1&1\end{smallmatrix}$, i.e., if its evaluation under v is negative. The conclusion of Proposition 4.2 can be rephrased by saying that the number of negative cells in ℓ and the number of polygons in P have the same parity, i.e., are both even or both odd. For example, ℓ_0 satisfies the proposition because ℓ_0 has no negative cells and P_0 has no polygons.

Assume now that some P_{i-1} and ℓ_{i-1} satisfy the proposition, and let us show that the same is true for P_i and ℓ_i. This will complete the proposition. We know that we are in one of the twelve cases of Figures 6 and 7, and we must look at the number of negative cells in ℓ_i versus ℓ_{i-1}, and the number of polygons in P_i versus P_{i-1}. An easy case is (a) of

Figure 7 because ℓ_i has one more negative cell than ℓ_{i-1} (specifically $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$), and P_i has one more polygon than P_{i-1}.

In the remaining cases let us now look at how the number of negative cells changes from ℓ_{i-1} to ℓ_i. In the six cases of Figure 6 and also case (d) of Figure 7, the four pictured cells have no negative cells regardless of whether # equals 0 or 1, so there is no change. The number is also unchanged in (b) of Figure 7 because a single negative cell $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$ changes position between ℓ_{i-1} and ℓ_i. In (c) of Figure 7, the number of negative cells increases by 2 because changing 0 to 1 in the central position creates both a cell $\begin{smallmatrix}1&0\\1&1\end{smallmatrix}$ and a cell $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$. Finally, the number decreases by 1 in (e) of Figure 7 and increases by 1 in (f) of Figure 7. Therefore, we can complete the proposition by showing that in the six cases of Figure 6 and in (b), (c), (d) of Figure 7, P_{i-1} and P_i have the same number of polygons, whereas in (e) and (f) of Figure 7, the number of polygons in P_{i-1} and P_i differs by 1.

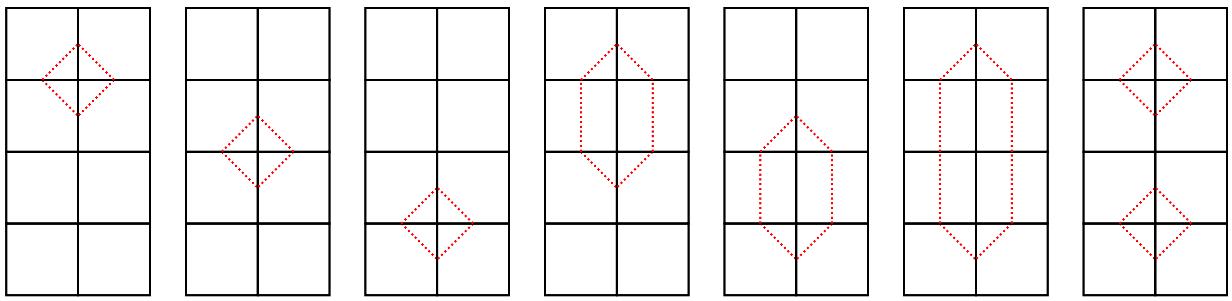
Consider, for example, (c) of Figure 6, and look just above the lower left corner and just left of the upper right corner. In P_{i-1} these are connected by a straight dashed line segment, and in P_i these are connected by dotted line segments. We call each of these an *inside connection*. Because P_{i-1} and P_i are polygon mosaics, these same points are connected by line segments outside the pictured area, and we will call this an *outside connection* because it connects points on the edge of the pictured area with a path outside the area. This outside connection is the same in P_{i-1} and P_i. Therefore P_{i-1} and P_i have the same number of polygons because the polygon with this outside connection changes shape, and all other polygons are unchanged. Similar reasoning completes the remaining cases of Figure 6 and cases (b), (c), (d) of Figure 7.

We now must show that in (e) and (f) of Figure 7, the number of polygons in P_{i-1} and P_i differ by 1. We will explicitly show this in (e), and essentially identical reasoning applies to (f). There are four places where inside connections meet the edge of the pictured area. Beginning in the lower left and working clockwise, let x_1, x_2, x_3, and x_4 be these points. For example, x_3 is just left of the upper right corner and x_4 is just above the lower right corner. In P_{i-1}, there is an inside connection between x_1 and x_2, and another between x_3 and x_4, whereas P_i has inside connections between x_1 and x_4, and between x_2 and x_3.

Any outside connections in P_{i-1} and P_i must be identical, and since these are polygon mosaics, each of x_1, x_2, x_3, and x_4 is the end of some outside connection. One possibility is that we have one outside connection between x_1 and x_2, and another between x_3 and x_4. In this case, the inside and outside connections of P_{i-1} complete two polygons, one including x_1 and x_2, and the other including x_3 and x_4. However, P_i will have one fewer polygon because a single polygon contains all x_i. Another possibility is to have one outside connection between x_1 and x_4, and another between x_2 and x_3. In this case, in P_{i-1} a single polygon includes all x_i, whereas in P_i they complete two polygons, one containing x_1 and x_4, and the other containing x_2 and x_3, and therefore P_i has one more polygon than P_{i-1}. These are the only possibilities since it is impossible to have an outside connection between x_1 and x_3, and another between x_2 and x_4. This is because our polygons are in a plane, and polygons in a mosaic are not able to cross one another. We have completed our proof. $\square$

Proof of Theorem 4.1. Fixing m and n, throughout this proof we use "polygon mosaic" for "(m, n) polygon mosaic," "framed lattice" for "(m, n) framed lattice," and so on.

We introduce the following identification for polygon mosaics. Let $\{P_i\}_{i \in J}$ be the set of all polygon mosaics that have exactly one polygon, and for any subset I of J, let P_I, if it exists, be the polygon mosaic that contains exactly the polygons of P_i for all $i \in I$. For example, $P_\varnothing$ is the polygon mosaic containing no polygons, i.e., the mosaic entirely composed of blank tiles, and for any $i \in J$, $P_{\{i\}}$ always exists and equals P_i. For an arbitrary $I \subseteq J$, P_I exists if and only if the polygons of P_i, for $i \in I$, do not "overlap." More precisely, P_I exists if and only if for every mosaic location, there is at most one $i \in I$ such that P_i has a non-blank tile there. Notice that every polygon mosaic can be uniquely written as P_I for some $I \subseteq J$. For example, when $(m, n) = (4, 2)$, we can take $J = \{1, 2, 3, 4, 5, 6\}$ and P_I as in Figure 8.



$P_{\{1\}}$ $P_{\{2\}}$ $P_{\{3\}}$ $P_{\{4\}}$ $P_{\{5\}}$ $P_{\{6\}}$ $P_{\{1,3\}}$

Figure 8: All $(4, 2)$ polygon mosaics except $P_\varnothing$

Next we look at some sets of mosaics. For any $I \subseteq J$ such that P_I exists, let M_I be the set of mosaics that contain the polygons of P_I, i.e., the set of mosaics that can be formed from P_I by using any tile wherever P_I has a blank tile. Notice that the size $|M_I|$ of this set equals 7^a, where a is the number of blank tiles in P_I. By the definition of $\mathscr{L}$, the number of cells of $\mathscr{L}(P_I)$ that are either $\begin{smallmatrix}0&0\\0&0\end{smallmatrix}$ or $\begin{smallmatrix}1&1\\1&1\end{smallmatrix}$ is also a, and since $\mathscr{L}(P_I)$ is proper, and thus it has neither $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$ nor $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$ cells, we see that $|v(\mathscr{L}(P_I))| = |M_I|$. Furthermore, $|I|$ is the number of polygons in P_I, so Proposition 4.2 tells us that $v(\mathscr{L}(P_I))$ and $(-1)^{|I|}$ have the same sign. Therefore, if P_I exists then

$$v(\mathscr{L}(P_I)) = (-1)^{|I|} |M_I| .$$

We must show that the sum $\sum_\ell v(\ell)$ over all framed lattices ℓ is equal to the number of polygon free mosaics. The sum is unchanged if we restrict the sum to all proper lattices ℓ because $v(\ell) = 0$ if ℓ is framed but not proper. Furthermore, since $\mathscr{L}$ gives a bijection from polygon mosaics to proper lattices,

$$\sum_\ell v(\ell) = \sum v(\mathscr{L}(P_I)) = \sum (-1)^{|I|} |M_I| ,$$

where the second and third sums are over all $I \subseteq J$ such that P_I exists.

The total number of mosaics, which happens to be 7^{mn}, equals $|M_\varnothing|$. Notice that $J = \varnothing$ if and only if $m = 1$ or $n = 1$. If this is the case then every mosaic is polygon free, the third sum above is $|M_\varnothing|$, and our theorem is complete. Therefore we may assume that $m \geq 2$ and $n \geq 2$, so that $J \neq \varnothing$ and the third sum has a nonzero term with $I \neq \varnothing$, for example, $I = \{i\}$ for any $i \in J$.

For any $I \neq \varnothing$ such that P_I exists, $M_I = \bigcap_{i \in I} M_{\{i\}}$ by the definition of M_I and the $M_{\{i\}}$. Thus

$$\sum_{\ell} v(\ell) = |M_{\varnothing}| - \sum (-1)^{|I|-1} \left| \bigcap_{i \in I} M_{\{i\}} \right|,$$

where the sum on the right is over all I such that P_I exists and $\varnothing \neq I \subseteq J$. In this equation, the value on the right side is unchanged if instead we sum over all I such that $\varnothing \neq I \subseteq J$. This is because when P_I is not defined, the P_i, $i \in I$, "overlap" in the sense described above, and so $\bigcap_{i \in I} M_{\{i\}} = \varnothing$. By the inclusion-exclusion principle (see the Appendix), this summation over all I with $\varnothing \neq I \subseteq J$ equals $\left|\bigcup_{i \in J} M_{\{i\}}\right|$, i.e., the number of mosaics that contain at least one polygon. Since $|M_{\varnothing}|$ is the total number of mosaics, we conclude, as desired, that $\sum_{\ell} v(\ell)$ is the number of polygon free mosaics. $\square$

Combining Theorem 4.1 with Theorem 3.2 gives the second main result of our paper.

Theorem 4.3. *Define the 1×1 matrices $A_1 = \begin{pmatrix}7\end{pmatrix}, B_1 = \begin{pmatrix}-1\end{pmatrix}$, and $C_1 = D_1 = \begin{pmatrix}1\end{pmatrix}$, and for positive integers n, define the $2^n \times 2^n$ matrices*

$$A_{n+1} = \begin{pmatrix} 7A_n & B_n \\ C_n & D_n \end{pmatrix} \quad B_{n+1} = \begin{pmatrix} -A_n & B_n \\ 0C_n & D_n \end{pmatrix}$$
$$C_{n+1} = \begin{pmatrix} A_n & 0B_n \\ C_n & D_n \end{pmatrix} \quad D_{n+1} = \begin{pmatrix} A_n & -B_n \\ C_n & 7D_n \end{pmatrix}.$$

Then for all positive integers m and n, the number of (m, n) polygon free mosaics is the upper left entry of A_n^m.

5 Related Work

Hong and Oh [4] provided bounds on the number of polygon mosaics.

Theorem 5.1 ([4])**.** *The number of (m, n) polygon mosaics for $m, n \geq 3$ is between $2^{m+n-3}\left(\frac{17}{10}\right)^{(m-2)(n-2)}$ and $2^{m+n-3}\left(\frac{31}{16}\right)^{(m-2)(n-2)}$.*

Our Theorem 3.4 compliments this result with a procedure to exactly count polygon mosaics, as well as recreate the sequence A181245 in the OEIS [6]. Note that unlike here, Hong and Oh did not consider the mosaic made from only blank tiles a polygon mosaic, and so our result differs from theirs by 1.

The methods in [11] could be used to count polygon mosaics, a fact which is likely known to the authors of [11], though to our knowledge has not appeared in print. However, their methods are incapable of enumerating our newly introduced polygon free mosaics. Our work extends their methods so we can count both polygon mosaics and polygon free mosaics.

In related work, Lomonaco and Kauffman [5] introduced mosaics constructed from the eleven distinct tile types of Figure 9, the first seven of which form our tile set.
The authors were interested in certain mosaics which they call knot mosaics, since the line segments can form knots. Analogous to polygon mosaics, *knot mosaics* are mosaics in which any tile that is not part of a knot is the blank tile. Oh et al. [11] counted the number of knot

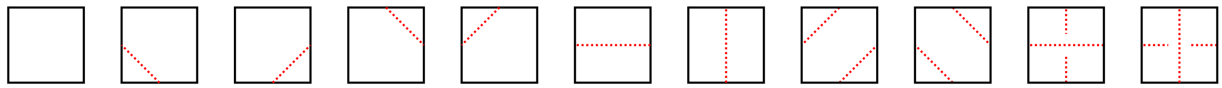


Figure 9: The Lomonaco and Kauffman tile set [5]

mosaics using an analagous method to that of Theorem 3.2. Oh and colleagues go beyond counting by also bounding the growth rate of knot mosaics [1, 8, 10], and Oh further adapts techniques similar to Theorem 3.2 to solve problems in monomer and dimer tilings [7, 9]. Related ideas were independently used to count the number of rectangular partitions of a rectangle in [12] for the sequence A182275 in the OEIS [6].

6 Extensions and Further Questions

It is simple to adapt our techniques to count polygon mosaics and polygon free mosaics in certain extensions of our tile set. Throughout this section, fix positive integers m and n, and let $\sum_\ell v(\ell)$ always designate the sum of $v(\ell)$ over all (m, n) framed lattices ℓ. In all cases, v is a cell product function, and therefore Theorem 3.2 tells us that matrices can be used to evaluate these sums.

Fewer tiles. If we limit ourselves to a subset of the tiles in Figure 1 then, of course, there are fewer polygon mosaics and polygon free mosaics. We can count how many by modifying our existing work. For example, omitting the rightmost tile of Figure 1, the number of polygon mosaics equals $\sum_\ell v(\ell)$, where v is the cell product function that maps $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$, $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$, $\begin{smallmatrix}0&1\\0&1\end{smallmatrix}$, and $\begin{smallmatrix}1&0\\1&0\end{smallmatrix}$ to 0 and the remaining twelve cells to 1. This can be shown using the reasoning of Theorem 3.1, since the omitted tile is the one that maps via $\mathscr{L}$ to either $\begin{smallmatrix}0&1\\0&1\end{smallmatrix}$ or $\begin{smallmatrix}1&0\\1&0\end{smallmatrix}$.

We can similarly count the number of polygon mosaics that can be formed by omitting a different tile or by omitting several tiles. It is worth noting, however, that some of these turn out to be trivial. Specifically, if the second, third, fourth, or fifth tile from left is omitted from Figure 1 then there is only one possible polygon mosaic, namely the one that contains only blank tiles. We outline three ways to see this when omitting the second tile; omitting the third, fourth, or fifth tile from left is similar. First assume a polygon mosaic has tiles that are not blank. Of these tiles, look at those in the highest possible row. The one farthest to the right must be the second tile of Figure 1. Second, in any framed binary lattice in which not all vertices are zero, among the vertices that are 1 and are as high as possible, the one farthest to the right must be the corner of a cell that is $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$. Finally, leaving the details to the reader, in Theorem 3.2, if $v\left(\begin{smallmatrix}0&0\\1&0\end{smallmatrix}\right) = 0$ and $v\left(\begin{smallmatrix}0&0\\0&0\end{smallmatrix}\right) = 1$ then for all positive integers m and n, the upper right entry of A_n^m equals 1.

As for determining the number of polygon free (m, n) mosaics using a reduced tile set, this will simply be 7^{mn} if the second, third, fourth, or fifth tile is omitted from the tile set, since we have just seen that no polygons can be formed. For these or other omitted tiles, we can use the ideas from Theorem 4.1. For example, if we omit the last tile of Figure 1 then we use the cell product function v that maps $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$, $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$, $\begin{smallmatrix}0&1\\0&1\end{smallmatrix}$, and $\begin{smallmatrix}1&0\\1&0\end{smallmatrix}$ to 0, $\begin{smallmatrix}0&0\\0&0\end{smallmatrix}$ and $\begin{smallmatrix}1&1\\1&1\end{smallmatrix}$ to 6 (since there are six tiles that can replace the blank tile), $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$ and $\begin{smallmatrix}1&0\\1&1\end{smallmatrix}$ to -1, and the remaining eight cells to 1. For omitting the blank tile instead, we would use v defined by mapping $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$ and $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$ to 0, $\begin{smallmatrix}0&0\\0&0\end{smallmatrix}$ and $\begin{smallmatrix}1&1\\1&1\end{smallmatrix}$ to 6, $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$ and $\begin{smallmatrix}1&0\\1&1\end{smallmatrix}$ to -1, and the remaining ten tiles to 1. In this case we would consider polygon mosaics that might include blank tiles, and create only those polygon free mosaics in which the blank tile was replaced by a tile that is not blank. This problem was posed by the first author of this paper and served as the motivation for all our work. The number of polygon free mosaics with the blank tile omitted is A364440 in the OEIS [6].

Adding tile variations. Adding tiles with the same connection points is similar. For instance, consider including three types of tiles that connect the top and bottom midpoints such as in Figure 10.

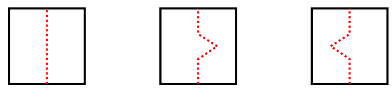


Figure 10: Various tiles that connect the top and bottom midpoints

To count polygon mosaics in this new tile set, take v such that $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$ and $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$ map to 0, $\begin{smallmatrix}0&1\\0&1\end{smallmatrix}$ and $\begin{smallmatrix}1&0\\1&0\end{smallmatrix}$ map to 3, and all other cells map to 1. To count polygon free mosaics, take v that maps $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$ and $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$ to 0, $\begin{smallmatrix}0&1\\0&1\end{smallmatrix}$ and $\begin{smallmatrix}1&0\\1&0\end{smallmatrix}$ to 3, $\begin{smallmatrix}0&0\\0&0\end{smallmatrix}$ and $\begin{smallmatrix}1&1\\1&1\end{smallmatrix}$ to 9, $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$ and $\begin{smallmatrix}1&0\\1&1\end{smallmatrix}$ to -1, and map the eight remaining cells to 1.

One could also construct a tile set with both styles of tiles in Figure 3. If this is done for each tile that joins adjacent sides of the unit square, then to count polygon mosaics take v such that $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$ and $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$ map to 0, $\begin{smallmatrix}0&1\\0&1\end{smallmatrix}$, $\begin{smallmatrix}1&0\\1&0\end{smallmatrix}$, $\begin{smallmatrix}0&0\\1&1\end{smallmatrix}$, $\begin{smallmatrix}1&1\\0&0\end{smallmatrix}$, $\begin{smallmatrix}0&0\\0&0\end{smallmatrix}$, and $\begin{smallmatrix}1&1\\1&1\end{smallmatrix}$ map to 1, and all other cells map to 2. To count polygon free mosaics, take v that maps $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$ and $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$ to 0, $\begin{smallmatrix}0&1\\0&1\end{smallmatrix}$, $\begin{smallmatrix}1&0\\1&0\end{smallmatrix}$, $\begin{smallmatrix}0&0\\1&1\end{smallmatrix}$, and $\begin{smallmatrix}1&1\\0&0\end{smallmatrix}$ map to 1, $\begin{smallmatrix}0&0\\0&0\end{smallmatrix}$ and $\begin{smallmatrix}1&1\\1&1\end{smallmatrix}$ to 11, $\begin{smallmatrix}0&0\\1&0\end{smallmatrix}$ and $\begin{smallmatrix}1&0\\1&1\end{smallmatrix}$ to -2, and map the six remaining cells to 2.

The Lomonaco and Kauffman tile set. We can count knot mosaics by taking v such that $\begin{smallmatrix}0&1\\1&0\end{smallmatrix}$ and $\begin{smallmatrix}1&0\\0&1\end{smallmatrix}$ map to 4, and map the fourteen remaining cells to 1. This is equivalent to the main result in [11]. Although we could consider counting the number of knot free mosaics built from the Lomonaco and Kauffman tile set in Figure 9, we will not do so. For this tile set, there are two reasonable ways to define a knot free mosaic. A knot free mosaic could either be a mosaic in which no knot can be found, or one in which no set of tiles exists for which all segments in those tiles are part of knots. For example, the mosaic in Figure 11 is knot free in the second sense but not in the first.

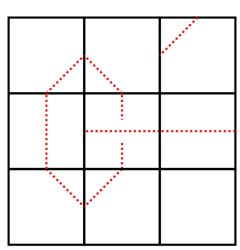


Figure 11: A mosaic that is knot free in the second sense but not the first

We invite others to find a way to count such mosaics, but we indicate a potential difficulty.

The end of the proof of Proposition 4.2 relies on polygons not being able to cross one another, something that is possible with knots formed from the Lomonaco and Kauffman tile set.

7 Acknowledgment

The authors would like to thank Richard Schank for contributing ideas toward the definition of $\mathscr{L}$ and ideas that led to Theorem 3.2.

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8 Appendix

Proposition (The inclusion-exclusion principle)**.** *For finite sets $A_1, A_2, \ldots, A_M$,*

$$\left|\bigcup_{j=1}^{M} A_j\right| = \sum_{\varnothing \neq I \subseteq \{1,2,\ldots,M\}} (-1)^{|I|-1} \left|\bigcap_{j\in I} A_j\right|.$$

Proof. The $M = 1$ case is trivial, and $M = 2$ case is the familiar and easily justified statement that $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$. We proceed inductively, so for $M \geq 3$, assume the result for $M - 1$ sets, and let us prove it for M sets. We split up the sum on the right of our desired equation into sets I that do not contain M and those that do. By our inductive hypothesis, the first sum equals $\left|\bigcup_{j=1}^{M-1} A_j\right|$. The second sum is $\sum_{\{M\}\subseteq I\subseteq\{1,2,\ldots,M\}} (-1)^{|I|-1} \left|\bigcap_{j\in I} A_j\right|$, which we rewrite by separating the $I = \{M\}$ term and reindexing the rest using $J = I \setminus \{M\}$ to get

$$|A_M| - \sum_{\varnothing \neq J \subseteq \{1,2,\ldots,M-1\}} (-1)^{|J|-1} \left|\bigcap_{j\in J} (A_j \cap A_M)\right|.$$

Again using our inductive hypothesis, this final sum equals $\left|\bigcup_{j=1}^{M-1} (A_j \cap A_M)\right| = \left|\left(\bigcup_{j=1}^{M-1} A_j\right) \cap A_M\right|$. The entire right side of our desired equation therefore equals $\left|\bigcup_{j=1}^{M-1} A_j\right| + |A_M| - \left|\left(\bigcup_{j=1}^{M-1} A_j\right) \cap A_M\right|$. By the $M = 2$ case, this is $\left|\bigcup_{j=1}^{M} A_j\right|$, as desired. $\square$